

Abhinna Manandhar

New York | abhinna@buffalo.edu | +1 (716) 247-9004 | Website | LinkedIn | GitHub

Education

University at Buffalo, State University of New York
Master of Science in Computer Science and Engineering

May 2026

- **GPA** 3.7/4.0
- **Coursework:** Algorithm Analysis and Design, Operating Systems, Intro to Machine Learning, Database Systems

Coventry University

November 2023

Bachelor of Science with Honours (BSc. Hons) in Computing

- **Grade:** First-Class Honours
- **Thesis:** Blockchain Technology to Improve Security and Accessibility of Electronic Health Records

Skills

Programming Languages: Python, Java, C, C++ TypeScript/JavaScript, Dart

Frameworks & Tools: REST, gRPC, GraphQL, FastAPI, Django, PyTest, Git, JIRA

Databases: PostgreSQL, MySQL, MongoDB

Systems: Linux/Unix, Distributed Systems, Docker, Redis, AWS S3, CI/CD, GitHub Actions, Grafana

ML & Data: Pandas, Matplotlib, Scikit-Learn, TensorFlow, Keras, PyTorch, LLM/GenAI Integration

Experience

Research Assistant, Dept. of Earth Sciences – University at Buffalo

Apr 2025 – Jan 2026

- Trained a YOLOv8 model for detecting water-gauge ROIs from crowd-sourced images, and implemented an ML pipeline to visually label water-line and extract water-level values using OpenCV and Google Gemini API.
- Reduced API latency by designing and implementing an asynchronous service layer using *Redis queue* to execute the ML pipeline.
- Improved operational reliability and compliance with UB IT security standards by orchestrating regular Python package version upgrades and legacy server and database migration to a new UB *Linux* server.

Mid-Level Software Engineer, Krispcall – Kathmandu, Nepal

Aug 2024 – Jan 2025

- Led the development of the billing microservice to decouple the payment workflows from the core monolith, serving *10,000+* users and enabling *\$3M+* in ARR.
- Minimized custom logic to improve the system scalability by re-architecting the legacy billing system by adopting native Stripe primitives (Payment Intents, Subscriptions, Metered Billing) and designing normalized *PostgreSQL* schemas.
- Improved code quality and test coverage by maintaining *PyTest* test suites and enforcing *SonarQube* quality gates in the *CI/CD* pipeline.
- Implemented a Slack-based monitoring and alerting pipeline for sales and support operations, cutting support resolution time by 40%.

Associate Software Engineer, Krispcall – Kathmandu, Nepal

March 2023 – Aug 2024

- Implemented high-availability telephony features (Call Barging, Call Whispering, SMS Forwarding) to support 50,000+ minutes in daily call traffic using *Twilio*.
- Developed and maintained *GraphQL* and *gRPC APIs* for client and inter-service communication in *Python* and *FastAPI*, implementing *Repository* and *Unit-Of-Work design patterns* for data access abstraction and transaction management.
- Maintained *PostgreSQL* database migrations using *Alembic* and wrote optimized queries to minimize database overhead using *SQLAlchemy ORM*.

Projects

UB Tycoon – MLH UB Hacking 2025 Best Use of Gemini API Winner

- Built an AI-powered semester gamification application using the Gemini API to parse syllabus PDFs and power an academic-advisor chatbot.
- Designed a MongoDB schema for storing game state, and implemented a NodeJS REST API integrated with a React client.

Ride Sharing Web & Mobile Application

- Developed a full-stack ride-sharing application, implementing the back-end using NodeJS and Express to build a REST API, and MongoDB to store application data.
- Integrated Google Maps Platform Services (Places, Directions, Distance Matrix) to calculate pricing and implement navigation.
- Developed a multi-platform client using Flutter (Android/iOS/WearOS) and ReactJS for web dashboard.

x86 Kernel Scheduler & User Programs (PintOS)

- Implemented a priority scheduler along with priority donation to mitigate priority inversion in C for the x86 PintOS kernel.
- Enabled user program execution by implementing stack-based argument passing and core system call handlers such as *exec*, *wait*, and *file I/O*.